

Category Theory and Functional Programming

The Curry-Howard Correspondence

Andrés Sicard-Ramírez

Universidad EAFIT

Semestre 2022-2

Preliminaries

Textbook

S. Abramsky and N. Tzevelekos (2011). Introduction to Categories and Categorical Logic.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Outline

Typed Lambda-Calculus

References

Outline

Typed Lambda-Calculus

References

Typed Lambda-Calculus

Definition

A **typed λ -calculus** with function types, product types, unit type and empty type, is defined by:

(i) Types

$\sigma, \tau ::= b$	(base types)
$\sigma \rightarrow \tau$	(function types)
$\sigma \times \tau$	(product types)
1	(unit type)
0	(empty type)

(continued on next slide)

(ii) Terms

$M, N ::= x$	(variable)
$\lambda x.M$	(λ -abstraction)
$M N$	(application)
$\langle M, N \rangle$	(pair)
$\pi_1 M \mid \pi_2 M$	(projections)
$\langle \rangle$	(unit)
absurd M	(absurd)

(continued on next slide)

(iv) Conversion rules

$$\lambda x.M = \lambda y.M[x := y] \quad (\alpha\text{-conversion})$$

$$(\lambda x.M) N = M[x := N] \quad (\beta\text{-conversion})$$

$$\lambda x.M x = M, \text{ with } x \text{ not in } M \quad (\eta\text{-conversion})$$

$$\pi_i \langle M_1, M_2 \rangle = M_i$$

$$M = \langle \pi_1 M, \pi_2 M \rangle$$

(continued on next slide)

(iii) Contexts

$\Gamma ::= \emptyset$

(empty context)

| $\Gamma, x : \sigma$, with x not in Γ

(context extension)

(continued on next slide)

Typed Lambda-Calculus

(v) Typing rules

- Hypothesis

$$\frac{}{\Gamma, x : \sigma \vdash x : \sigma} \text{hyp} \quad (\text{if } x : \sigma \in \Gamma)$$

- Function types

$$\frac{\Gamma, x : \sigma \vdash M : \tau}{\Gamma \vdash \lambda x. M : \sigma \rightarrow \tau} \rightarrow\text{I}$$

$$\frac{\Gamma \vdash M : \sigma \rightarrow \tau \quad \Gamma \vdash N : \sigma}{\Gamma \vdash M N : \tau} \rightarrow\text{E}$$

- Product types

$$\frac{\Gamma \vdash M : \sigma \quad \Gamma \vdash N : \tau}{\Gamma \vdash \langle M, N \rangle : \sigma \times \tau} \times\text{I}$$

$$\frac{\Gamma \vdash M : \sigma_1 \times \sigma_2}{\Gamma \vdash \pi_i M : \sigma_i} \times\text{E}$$

(continued on next slide)

Typed Lambda-Calculus

(v) Typing rules

- Unit type

$$\frac{}{\Gamma \vdash \langle \rangle : \mathbf{1}} \text{1I}$$

- Empty type

$$\frac{\Gamma \vdash M : \mathbf{0}}{\Gamma \vdash \text{absurd } M : \sigma} \text{0E}$$

(continued on next slide)

Typed Lambda-Calculus

Category associated to the typed λ -calculus

The category **TLC** is defined by:

- (i) Objects: Types
- (ii) Arrows: There is an arrow $\sigma \xrightarrow{M} \tau$ iff $M : \sigma \rightarrow \tau$.

(continued on next slide)

(iii) Composition

If $M : \rho \rightarrow \sigma$ and $N : \sigma \rightarrow \tau$ then

$$M \circ N : \rho \rightarrow \tau$$

$$M \circ N := \lambda x. M (N x)$$

(iv) Identity arrows

$$\text{id}_\sigma : \sigma \rightarrow \sigma$$

$$\text{id}_\sigma := \lambda x. x, \text{ with } x : \sigma$$

(continued on next slide)

(v) Unit axiom

Let $M : \sigma \rightarrow \tau$, then

$$\begin{aligned} M \circ \text{id}_\sigma &= \lambda x. M(\text{id}_\sigma x) \\ &= \lambda x. M((\lambda y. y) x) \\ &= \lambda x. M x \\ &= M \end{aligned}$$

$$\begin{aligned} \text{id}_\tau \circ M &= \lambda x. \text{id}_\tau (M x) \\ &= \lambda x. (\lambda y. y) (M x) \\ &= \lambda x. M x \\ &= M \end{aligned}$$

(continued on next slide)

(vi) Associativity axiom

Let $M : \sigma_1 \rightarrow \sigma_2$, $N : \sigma_2 \rightarrow \sigma_3$ and $P : \sigma_3 \rightarrow \sigma_4$. Then

$$\begin{aligned} M \circ (N \circ P) &: \sigma_1 \rightarrow \sigma_4 \\ M \circ (N \circ P) &= \lambda x. M ((N \circ P) x) \\ &= \lambda x. M ((\lambda y. N (P y)) x) \\ &= \lambda x. M (N (P x)) \\ &= \lambda x. (\lambda y. M (N y)) (P x) \\ &= \lambda x. (M \circ N) (P x) \\ &= (M \circ N) \circ P \end{aligned}$$

(continued on next slide)

Typed Lambda-Calculus

(vii) Initial object: Type 0

$$i_\sigma : 0 \rightarrow \sigma$$

$$i_\sigma := \text{absurd}$$

(viii) Terminal object: Type 1

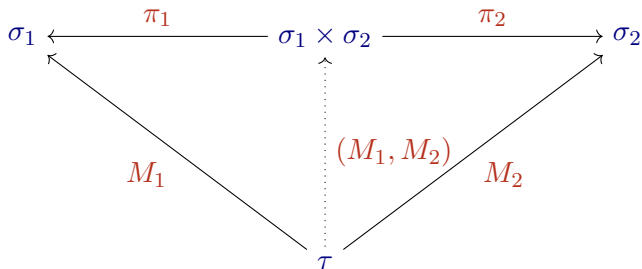
$$t_\sigma : \sigma \rightarrow 1$$

$$t_\sigma := \lambda x. \langle \rangle$$

(continued on next slide)

(ix) Binary products

We start with the objects and the arrows in the following diagram



where $(M_1, M_2) := \lambda x. \langle M_1 x, M_2 x \rangle : \tau \rightarrow \sigma_1 \times \sigma_2$,

(continued on next slide)

First, we prove that the diagram is commutative.

$$\begin{aligned}\pi_i \circ (M_1, M_2) &: \tau \rightarrow \sigma_i \\ \pi_i \circ (M_1, M_2) &= \pi_i \circ \lambda y. \langle M_1 y, M_2 y \rangle \\ &= \lambda x. \pi_i ((\lambda y. \langle M_1 y, M_2 y \rangle) x) \\ &= \lambda x. \pi_i \langle M_1 x, M_2 x \rangle \\ &= \lambda x. M_i x \\ &= M_i\end{aligned}$$

(continued on next slide)

Typed Lambda-Calculus

Next, we prove the uniqueness of arrow (M_1, M_2) . Let $N : \sigma_1 \times \sigma_2$ such that $\pi_i \circ N = M_i$. Then

$$\begin{aligned}(M_1, M_2) &= \lambda x. \langle M_1 x, M_2 x \rangle \\ &= \lambda x. \langle (\pi_1 \circ N) x, (\pi_2 \circ N) x \rangle \\ &= \lambda x. \langle (\lambda y. \pi_1 (N y)) x, (\lambda y. \pi_2 (N y)) x \rangle \\ &= \lambda x. \langle \pi_1 (N x), \pi_2 (N x) \rangle \\ &= \lambda x. N x \\ &= N\end{aligned}$$

(continued on next slide)

(x) Finitary products

Since **TLC** has binary products and a terminal object (nullary product) then it has finite products.

Outline

Typed Lambda-Calculus

References

References

- S. Abramsky and N. Tzevelekos (2011). Introduction to Categories and Categorical Logic. In: New Structures for Physics. Ed. by Bob Coecke. Vol. 813. Lecture Notes in Physics. Springer, pp. 3–94. DOI: [10.1007/978-3-642-12821-9_1](https://doi.org/10.1007/978-3-642-12821-9_1) (cit. on p. 2).